

Multivariate Random Variables

Joint Laws, Independence, Dependence, and the Multivariate Normal

Multivariate Random Variables

Definition — Random Vector

A **multivariate random variable** (or random vector) is a function

$$\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$$

where Ω is the sample space and $n \geq 2$.

We write $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$, where each X_i is a univariate random variable.

Examples

- (X, Y) — height and weight of a person
- (X_1, X_2, X_3) — temperature, humidity, pressure
- The stock prices of several companies at the close of trading

Joint Cumulative Distribution Function

Definition — Joint CDF

The **joint cumulative distribution function** of $\mathbf{X} = (X_1, \dots, X_n)$ is

$$F_{\mathbf{X}}(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Properties:

- (1) $0 \leq F_{\mathbf{X}}(x_1, \dots, x_n) \leq 1$
- (2) $F_{\mathbf{X}}$ is non-decreasing in **each** argument
- (3) $F_{\mathbf{X}}(-\infty, \dots, -\infty) = 0$ and $F_{\mathbf{X}}(\infty, \dots, \infty) = 1$
- (4) Right-continuous in each argument

Why the joint CDF matters

Exactly as in one dimension, it is the one object that always exists — discrete, continuous, or mixed.

Joint Probability Mass Function

For **discrete** random vectors $\mathbf{X} = (X_1, \dots, X_n)$:

i Definition — Joint PMF

$$p_{\mathbf{X}}(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Properties:

- (1) $p_{\mathbf{X}}(x_1, \dots, x_n) \geq 0$ for all $(x_1, \dots, x_n)$
- (2) $\sum_{x_1} \cdots \sum_{x_n} p_{\mathbf{X}}(x_1, \dots, x_n) = 1$
- (3) $F_{\mathbf{X}}(x_1, \dots, x_n) = \sum_{t_1 \leq x_1} \cdots \sum_{t_n \leq x_n} p_{\mathbf{X}}(t_1, \dots, t_n)$

Joint Probability Density Function

For **continuous** random vectors $\mathbf{X} = (X_1, \dots, X_n)$:

i Definition — Joint PDF

The joint density $f_{\mathbf{X}}$ satisfies

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{\mathbf{X}}(t_1, \dots, t_n) dt_n \cdots dt_1$$

Properties:

- (1) $f_{\mathbf{X}}(x_1, \dots, x_n) \geq 0$ for all $(x_1, \dots, x_n)$
- (2) $\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, \dots, x_n) dx_n \cdots dx_1 = 1$
- (3) $f_{\mathbf{X}}(x_1, \dots, x_n) = \frac{\partial^n F_{\mathbf{X}}(x_1, \dots, x_n)}{\partial x_1 \cdots \partial x_n}$

Marginals and Conditionals

Marginal Distributions

i Definition — Marginal Distribution

The **marginal distribution** of X_i is the distribution of X_i alone, obtained by summing or integrating the others away.

Discrete case (PMF):

$$p_{X_i}(x_i) = \sum_{x_1} \cdots \sum_{x_{i-1}} \sum_{x_{i+1}} \cdots \sum_{x_n} p_{\mathbf{X}}(x_1, \dots, x_n)$$

Continuous case (PDF):

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, \dots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n$$

Joint Marginal Distributions

For any **subset** of the variables, e.g. (X_1, X_2) from (X_1, X_2, X_3) :

Discrete:

$$p_{X_1, X_2}(x_1, x_2) = \sum_{x_3} p_{X_1, X_2, X_3}(x_1, x_2, x_3)$$

Continuous:

$$f_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{\infty} f_{X_1, X_2, X_3}(x_1, x_2, x_3) dx_3$$

! One-way street

The joint distribution determines **all** marginals. The marginals do **not** determine the joint — as the live demo two slides on will show.

Conditional Distributions

Given $\mathbf{X} = (X_1, \dots, X_k)$ and $\mathbf{Y} = (Y_1, \dots, Y_m)$:

i Conditional Densities

If $\mathbf{X}$ and $\mathbf{Y}$ are dependent, learning $\mathbf{Y} = \mathbf{y}$ updates the uncertainty about $\mathbf{X}$:

$$p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{p_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})}{p_{\mathbf{Y}}(\mathbf{y})} \quad \text{or} \quad f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})}{f_{\mathbf{Y}}(\mathbf{y})}$$

Properties:

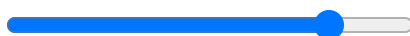
- Every property of a probability distribution still holds
- If $\mathbf{X}$ and $\mathbf{Y}$ are independent, $f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = f_{\mathbf{X}}(\mathbf{x})$ — conditioning tells you nothing

Joint, Marginal, Conditional, Live

The joint sits in the middle; the margins are what you get by *integrating out*, the slice is what you get by *conditioning*.

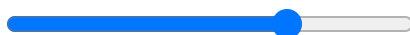
Correlation ρ

0.6



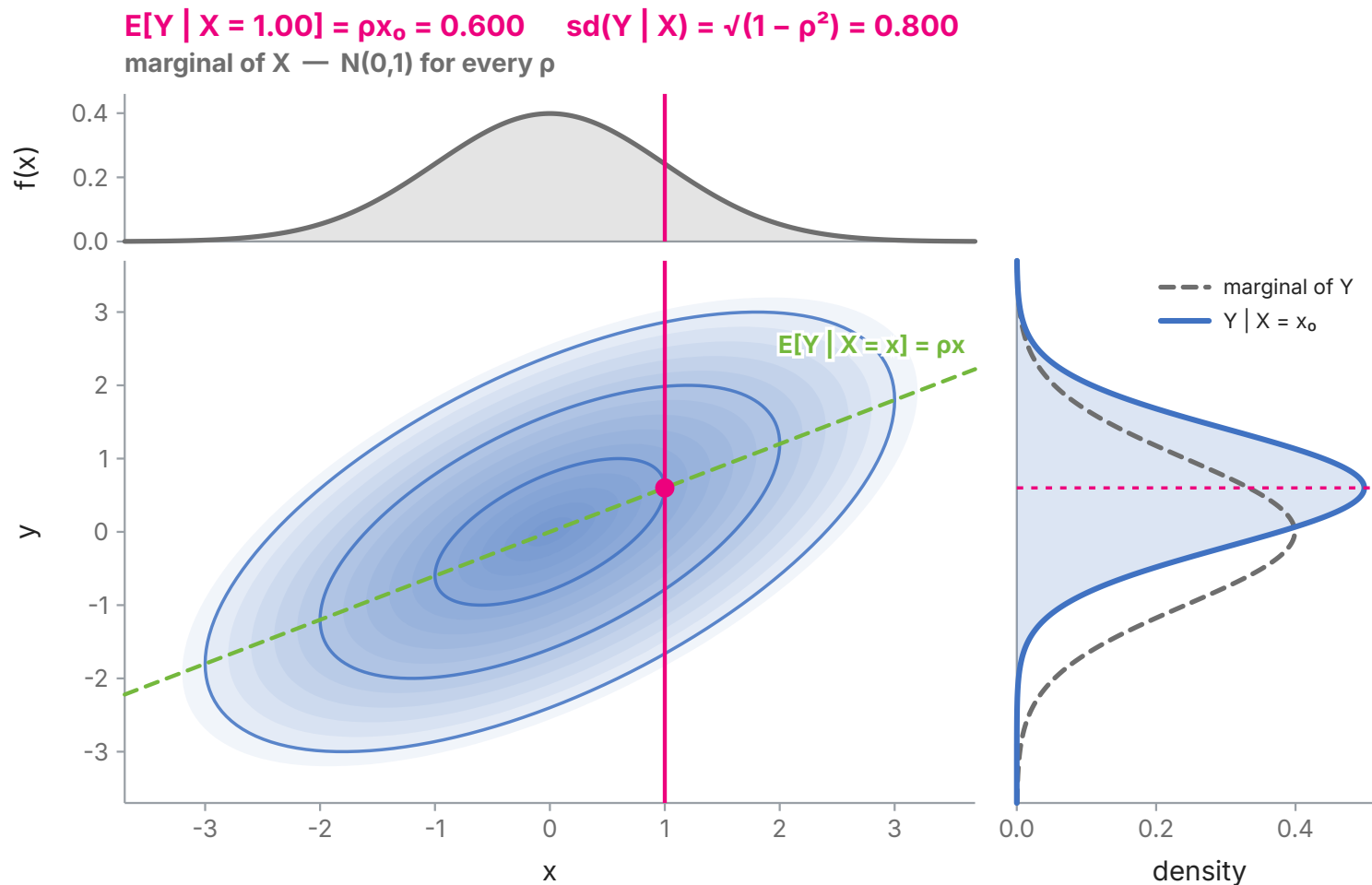
Condition on $X = x_0$

1



Standard bivariate normal. The top curve is the marginal of X ; the right panel shows the marginal of Y (grey) against the conditional $Y | X = x_0$ (blue).

The marginal of Y is $N(0, 1)$ whatever ρ is — only the joint changes.



Conditional Moments

Conditional Moments

$$E[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] = \int \mathbf{x} f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) d\mathbf{x}$$
$$\text{Var}[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] = E[\mathbf{X}\mathbf{X}^T \mid \mathbf{Y} = \mathbf{y}] - E[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] E[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}]^T$$

If $\mathbf{X}$ and $\mathbf{Y}$ are independent, then for every $\mathbf{y}$

- $E[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] = E[\mathbf{X}]$ and $\text{Var}[\mathbf{X} \mid \mathbf{Y} = \mathbf{y}] = \text{Var}[\mathbf{X}]$
- more generally $E[g(\mathbf{X}) \mid \mathbf{Y} = \mathbf{y}] = E[g(\mathbf{X})]$ for any g

Law of Total Expectation

$$E[\mathbf{X}] = E_{\mathbf{Y}}[E_{\mathbf{X}|\mathbf{Y}}[\mathbf{X} \mid \mathbf{Y}]] = \int_{\mathbf{Y}} \left[\int_{\mathbf{X}|\mathbf{Y}} \mathbf{x} f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) d\mathbf{x} \right] f_{\mathbf{Y}}(\mathbf{y}) d\mathbf{y}$$

Independence

Independence of Random Variables

i Definition — Mutual Independence

$X_1, \dots, X_n$ are **mutually independent** if

$$F_{\mathbf{X}}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdot F_{X_2}(x_2) \cdots F_{X_n}(x_n)$$

for all $(x_1, \dots, x_n) \in \mathbb{R}^n$.

Equivalent conditions:

- **Discrete:** $p_{\mathbf{X}}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$
- **Continuous:** $f_{\mathbf{X}}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$

Under independence the joint **is** the product of the marginals — the one case where the marginals do determine the joint.

Pairwise versus Mutual Independence

i Pairwise Independence

$X_1, \dots, X_n$ are **pairwise independent** if X_i and X_j are independent for every $i \neq j$.

i Mutual Independence

$X_1, \dots, X_n$ are **mutually independent** if **every** subset $\{X_{i_1}, \dots, X_{i_k}\}$ consists of independent variables.

! Key Point

Mutual independence is a **strictly stronger** condition.

Mutual independence $\implies$ Pairwise independence

Pairwise independence $\not\implies$ Mutual independence

Classic Example: Three Coin Tosses

Toss a fair coin three times and define

A : the first coin is heads

B : the second coin is heads

C : the first two coins agree

Sample space: $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, all equally likely.

- $A = \{HHH, HHT, HTH, HTT\}$, so $P(A) = 1/2$
- $B = \{HHH, HHT, THH, THT\}$, so $P(B) = 1/2$
- $C = \{HHH, HHT, TTH, TTT\}$, so $P(C) = 1/2$

Every **pair** multiplies correctly: $P(A \cap B) = P(A \cap C) = P(B \cap C) = 1/4$.

But Not Mutually Independent

All three at once:

$$A \cap B \cap C = \{HHH, HHT\}, \quad P(A \cap B \cap C) = \frac{2}{8} = \frac{1}{4}$$

But mutual independence would require

$$P(A \cap B \cap C) = P(A)P(B)P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

! Conclusion

Since $\frac{1}{4} \neq \frac{1}{8}$, the three events are **pairwise independent** but **not mutually independent**.

Pairwise versus Mutual, Live

Every pair checks out. The triple does not. Step through them.

Check

A $\cap$ B

Eight equally likely outcomes. A column is shaded in a row when that outcome belongs to the event. The bottom row is the intersection of everything you selected — count its squares and compare with the product of the probabilities.

$P(A \cap B) = 2/8 = 0.250$ $P(A) \cdot P(B) = 0.250$
the two agree — this collection is independent

	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
A								
B								
C								
$\cap$								

Independence: Summary

- **Pairwise independence:** every *pair* of variables is independent
- **Mutual independence:** every *subset* of variables is independent
- Mutual independence is strictly stronger than pairwise independence
- Knowing that variables are pairwise independent is **not** enough to conclude mutual independence



Where this bites

The distinction matters wherever a product of probabilities is taken over more than two things at once — likelihoods, hashing arguments, and independence assumptions in machine learning models.

Dependence

Dependence of Random Variables

Mean Dependence

$$E[Y \mid X = x] \neq E[Y]$$

The conditional expectation of Y depends on the value of X .

Quantile Dependence

$$Q_{Y|X=x}(\tau) \text{ depends on } x \text{ for some } \tau \in (0, 1)$$

The conditional quantiles move with X even when the conditional mean does not — dependence in spread or shape, not in level.

Tail Dependence

Extreme values occur together. For the upper tail,

$$\lambda_U = \lim_{\alpha \rightarrow 1^-} P(Y > Q_Y(\alpha) \mid X > Q_X(\alpha))$$

Dependence Characteristics

Ways Y can depend on X :

- **Linear:** $Y = aX + b + \varepsilon$ — captured by correlation
- **Monotonic:** Y increases (or decreases) with X , but not linearly
- **Nonlinear:** $Y = X^2$, $Y = \sin(X)$, U-shaped relationships
- **Conditional:** the dependence itself varies across the support

Popular alternative measures:

- **Pearson's r** — correlation coefficient, captures *linear* dependence
- **Spearman's ρ** — correlation of *ranks*, captures *monotonic* dependence
- **Kendall's τ** — based on concordance of pairs, also monotonic
- **Mutual information** — $I(X, Y) = 0$ if and only if independence

Marginals versus Dependence, Live

Two separate choices: the **dependence structure**, and the **marginals**.

Dependence structure

Independent

Marginals

Normal

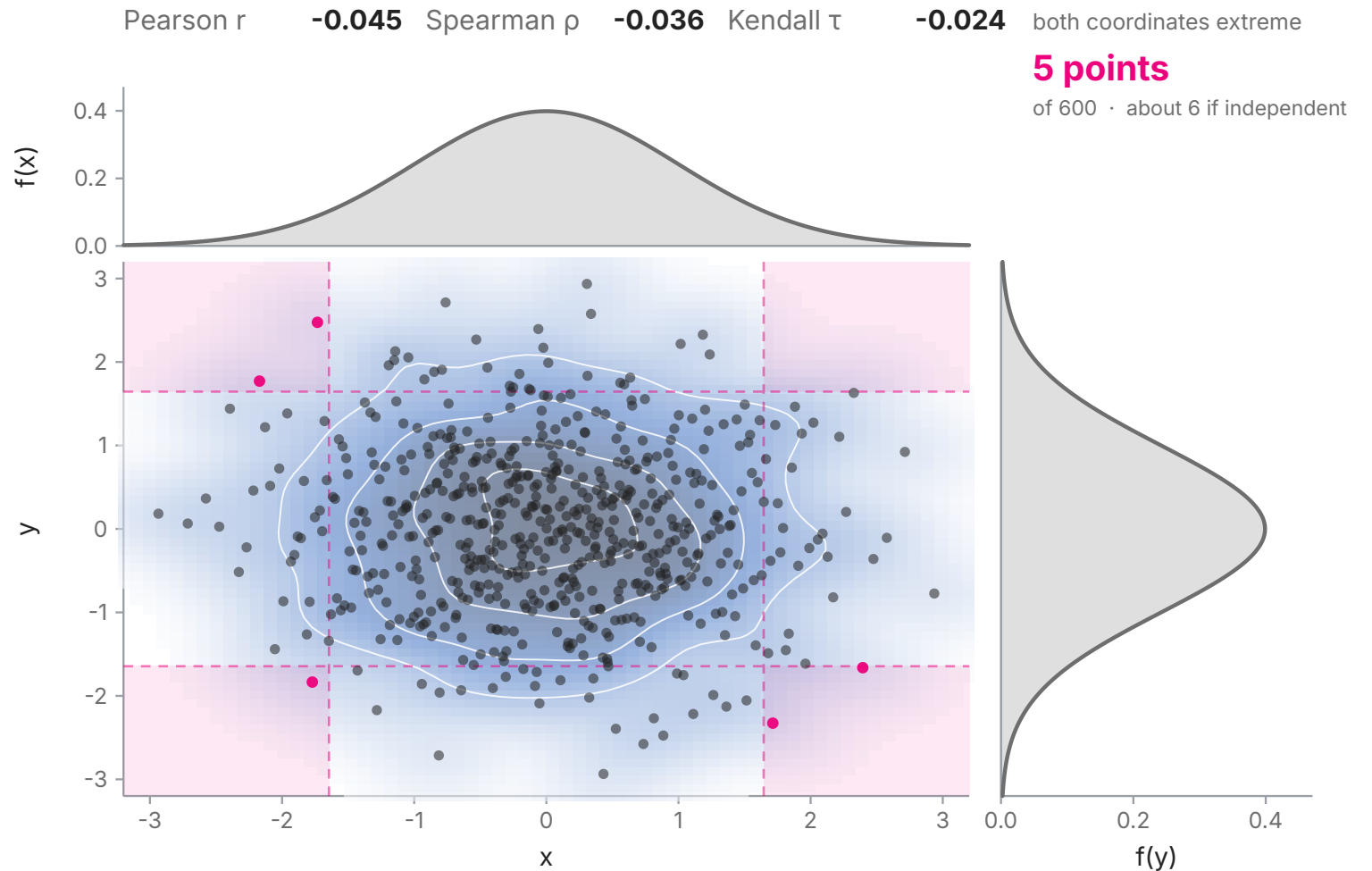
Show

Points + density

New sample

The structure sets the ranks — the copula.
The marginal only reshapes each axis.

U-shape and **tail-dependent**: all three measures read ≈ 0 , yet the pink corners are full. Switch to exponential marginals and Pearson jumps to $\approx +0.4$ — same ranks, different number.



Covariance and Correlation

Covariance

i Definition — Covariance

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Properties:

- $\text{Cov}(X, X) = \text{Var}(X)$ and $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$ — it is **not** scale-free
- If X and Y are independent then $\text{Cov}(X, Y) = 0$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$

! The converse fails

$\text{Cov}(X, Y) = 0$ does **not** imply independence — the U-shaped panel of the last demo is a counterexample.

Correlation

i Definition — Correlation Coefficient

$$\rho(X, Y) = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

provided $\text{Var}(X), \text{Var}(Y) > 0$.

Properties:

- $-1 \leq \rho(X, Y) \leq 1$
- $|\rho(X, Y)| = 1$ if and only if $Y = aX + b$ for constants $a \neq 0, b$
- $\rho(X, Y) = 0$ if X and Y are independent
- $\rho(aX + b, cY + d) = \text{sign}(ac) \cdot \rho(X, Y)$ — invariant up to sign

Covariance Matrix

For a random vector $\mathbf{X} = (X_1, \dots, X_n)^T$:

i Definition — Covariance Matrix

$$\mathbf{\Sigma} = \text{Cov}(\mathbf{X}) = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T], \quad \boldsymbol{\mu} = E[\mathbf{X}]$$

with (i, j) -th element $\Sigma_{ij} = \text{Cov}(X_i, X_j)$.

Properties:

- $\mathbf{\Sigma}$ is symmetric: $\mathbf{\Sigma} = \mathbf{\Sigma}^T$
- $\mathbf{\Sigma}$ is positive semidefinite
- Diagonal elements are variances: $\Sigma_{ii} = \text{Var}(X_i)$

The Covariance Ellipse, Live

Σ is an ellipse. Its axes are the eigenvectors; the axis lengths are $\sqrt{\lambda_i}$.

σ_x

1

σ_y

0.7

ρ

0.6

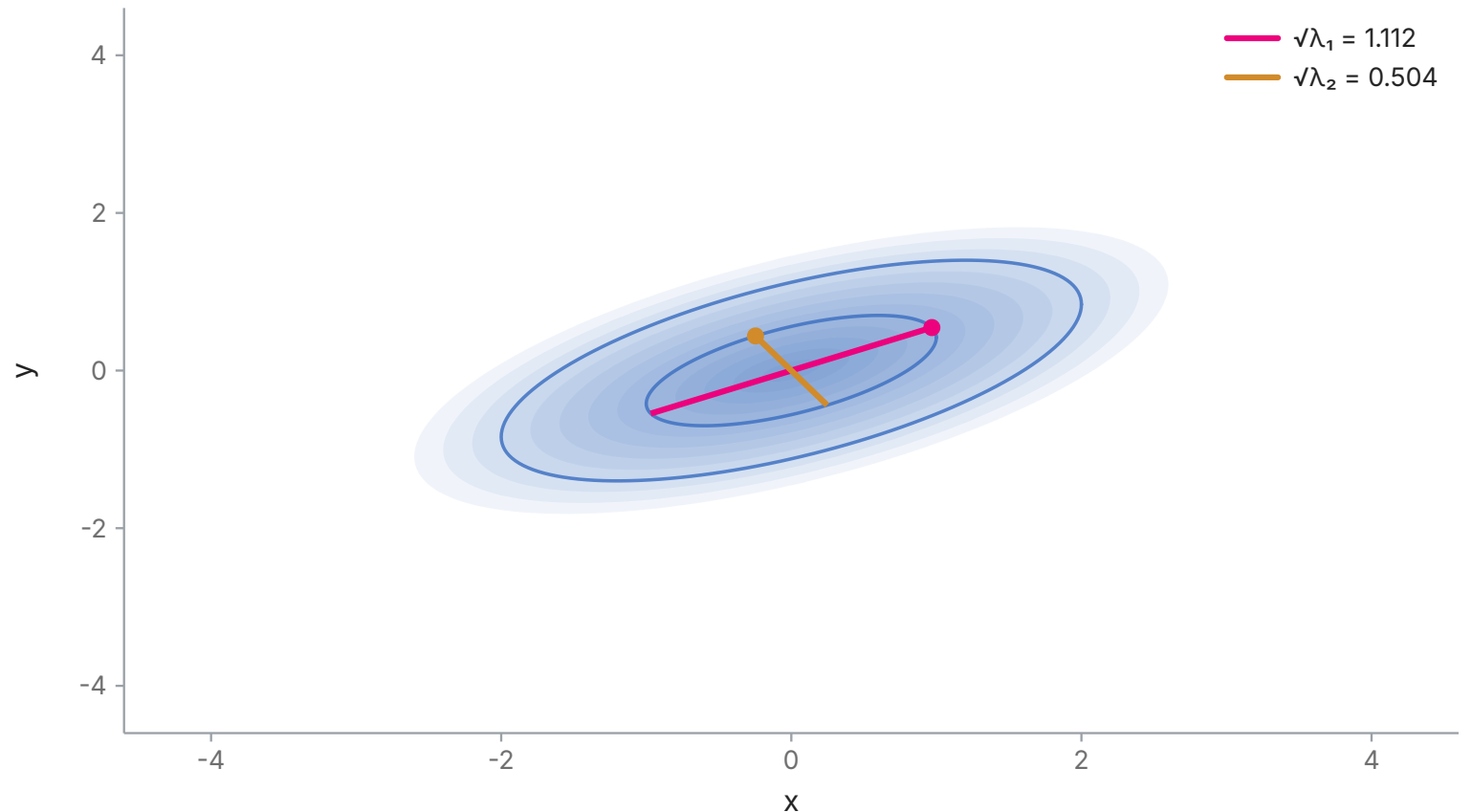
Re-scale X by c

1

Changing the *units* of $\mathbf{X}$ multiplies $\mathbf{Cov}$ by c but leaves ρ alone — that is exactly why correlation, not covariance, is the comparable number.

Push $|\rho| \rightarrow 1$ and the ellipse collapses onto a line: $\det \Sigma \rightarrow 0$, the boundary of positive semidefiniteness.

$\Sigma = [1.000, 0.420 ; 0.420, 0.490]$ $\det \Sigma = 0.3136$
 $\text{Cov}(X,Y) = 0.420$ — $\text{Corr}(X,Y) = 0.600$ (unchanged by c)



The Multivariate Normal

Multivariate Normal Distribution

i Definition — Multivariate Normal

$\mathbf{X} = (X_1, \dots, X_n)^T$ follows $\mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if its density is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

with $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\boldsymbol{\Sigma}$ positive definite.

Parameters: $\boldsymbol{\mu} = E[\mathbf{X}]$ (mean vector) and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{X})$ (covariance matrix).

Notation: $\mathbf{X} \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

The exponent is a quadratic form, so the level sets of the density are exactly the ellipses of the previous demo.

Properties of the Multivariate Normal

(1) **Linear combinations:** if $\mathbf{X} \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then

$$\mathbf{AX} + \mathbf{b} \sim \mathcal{N}_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T)$$

(2) **Marginals:** every marginal distribution is normal

(3) **Conditionals:** every conditional distribution is normal

(4) **Independence:** for the multivariate normal, uncorrelated $\Rightarrow$ independent

(5) **Quadratic forms:** $(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_n^2$

! Property 4 is special

In general zero correlation says nothing about independence. Inside the multivariate normal family it says everything.

Bivariate Normal Distribution

For $(X, Y) \sim \mathcal{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix}$$

the density is

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{Q}{2(1-\rho^2)}\right)$$

where $\rho = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$ and

$$Q = \frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}$$

Transformations

Transformations of Random Vectors

Given $\mathbf{X}$ with a known distribution, find the distribution of $\mathbf{Y} = \mathbf{g}(\mathbf{X})$, where $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

i Method 1 — CDF Method

$$F_{\mathbf{Y}}(\mathbf{y}) = P(\mathbf{g}(\mathbf{X}) \leq \mathbf{y}) = P(\mathbf{X} \in \{\mathbf{x} : \mathbf{g}(\mathbf{x}) \leq \mathbf{y}\})$$

i Method 2 — Jacobian Method (continuous)

If $\mathbf{g}$ is one-to-one with inverse $\mathbf{h}$, then

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{h}(\mathbf{y})) \cdot |J_{\mathbf{h}}(\mathbf{y})|$$

where $J_{\mathbf{h}}$ is the Jacobian determinant of $\mathbf{h}$.

Example: Linear Transformation

Problem. $\mathbf{X} \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, with $\mathbf{A}$ of size $m \times n$ and $\mathbf{b} \in \mathbb{R}^m$.

Solution.

$$\mathbf{Y} \sim \mathcal{N}_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T)$$

Special cases:

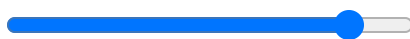
- **Sum:** $X + Y$ where (X, Y) is bivariate normal
- **Difference:** $X - Y$ where (X, Y) is bivariate normal
- **Sample mean:** $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ for i.i.d. normal X_i

Linear Transformations, Live

Rotate and stretch a normal cloud. It stays normal — only $\mathbf{A}\mathbf{\Sigma}\mathbf{A}^T$ changes.

Source ρ

0.7



Rotate A by (°)

0



Stretch first axis

1



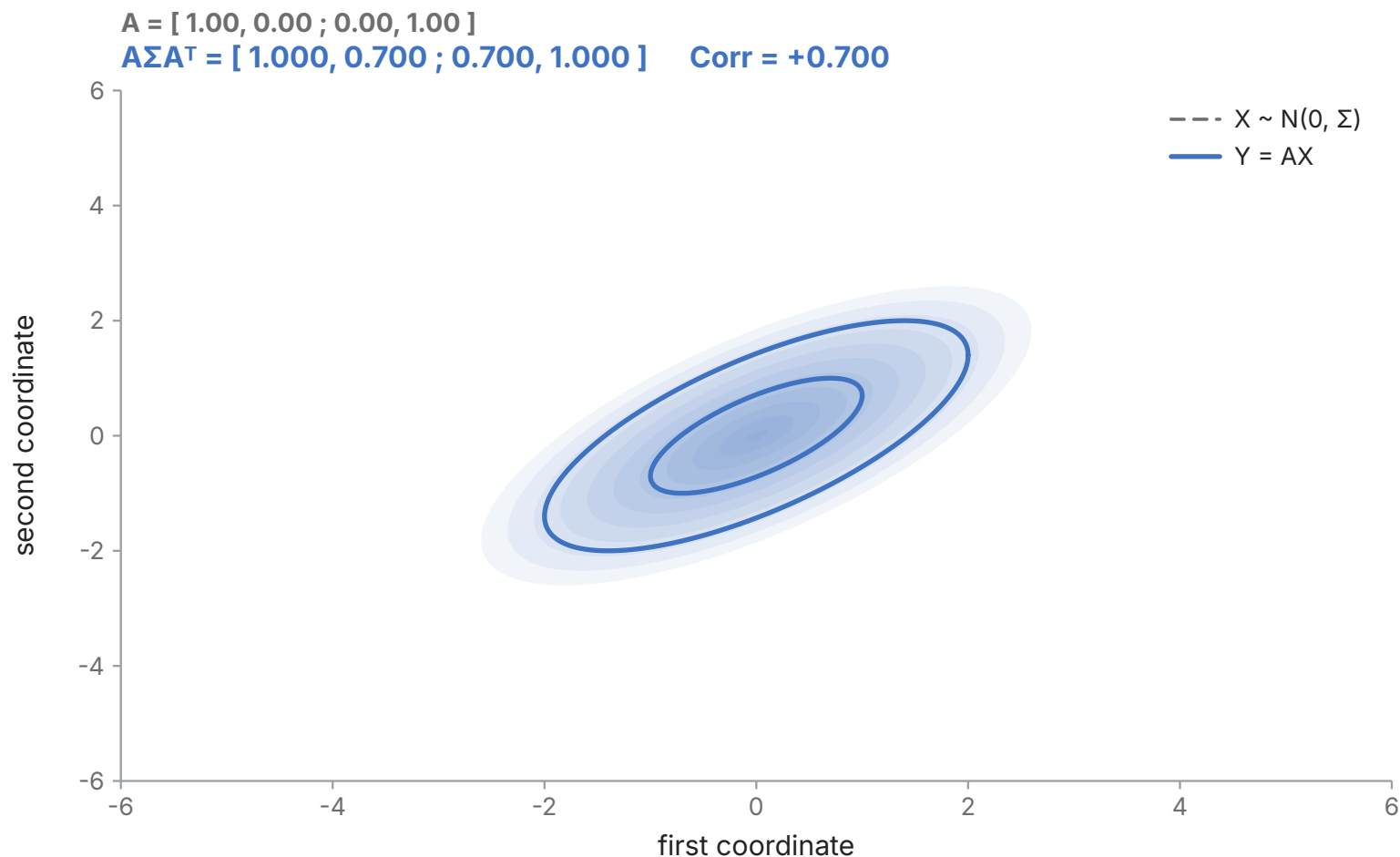
Stretch second axis

1



$\mathbf{A} = R(\theta) \text{diag}(a, b)$ applied to $\mathbf{X} \sim \mathcal{N}_2(\mathbf{0}, \mathbf{\Sigma})$ with unit variances and correlation ρ .

Set the rotation to 45° with equal stretches to see a sum-and-difference transform: $\mathbf{X} + \mathbf{Y}$ and $\mathbf{X} - \mathbf{Y}$ become **uncorrelated**.



Order Statistics

Order Statistics

Given n i.i.d. random variables $X_1, \dots, X_n$ with CDF F and density f :

i Definition — Order Statistics

The **order statistics** $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ are the sorted values of $X_1, \dots, X_n$.

Joint density of all order statistics:

$$f_{X_{(1)}, \dots, X_{(n)}}(x_1, \dots, x_n) = n! f(x_1) \cdots f(x_n), \quad x_1 \leq \dots \leq x_n$$

Density of the k -th order statistic:

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

Special Order Statistics

i Minimum

$$X_{(1)} = \min\{X_1, \dots, X_n\}$$

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$$

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

i Maximum

$$X_{(n)} = \max\{X_1, \dots, X_n\}$$

$$F_{X_{(n)}}(x) = [F(x)]^n$$

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x)$$

Range: $R = X_{(n)} - X_{(1)}$ — the spread of the whole sample, and itself a random variable.

Order Statistics, Live

n draws, sorted. Each faint curve is one order statistic; the bold one is the k -th.

Parent distribution

Uniform(0,1) ▼

Sample size n

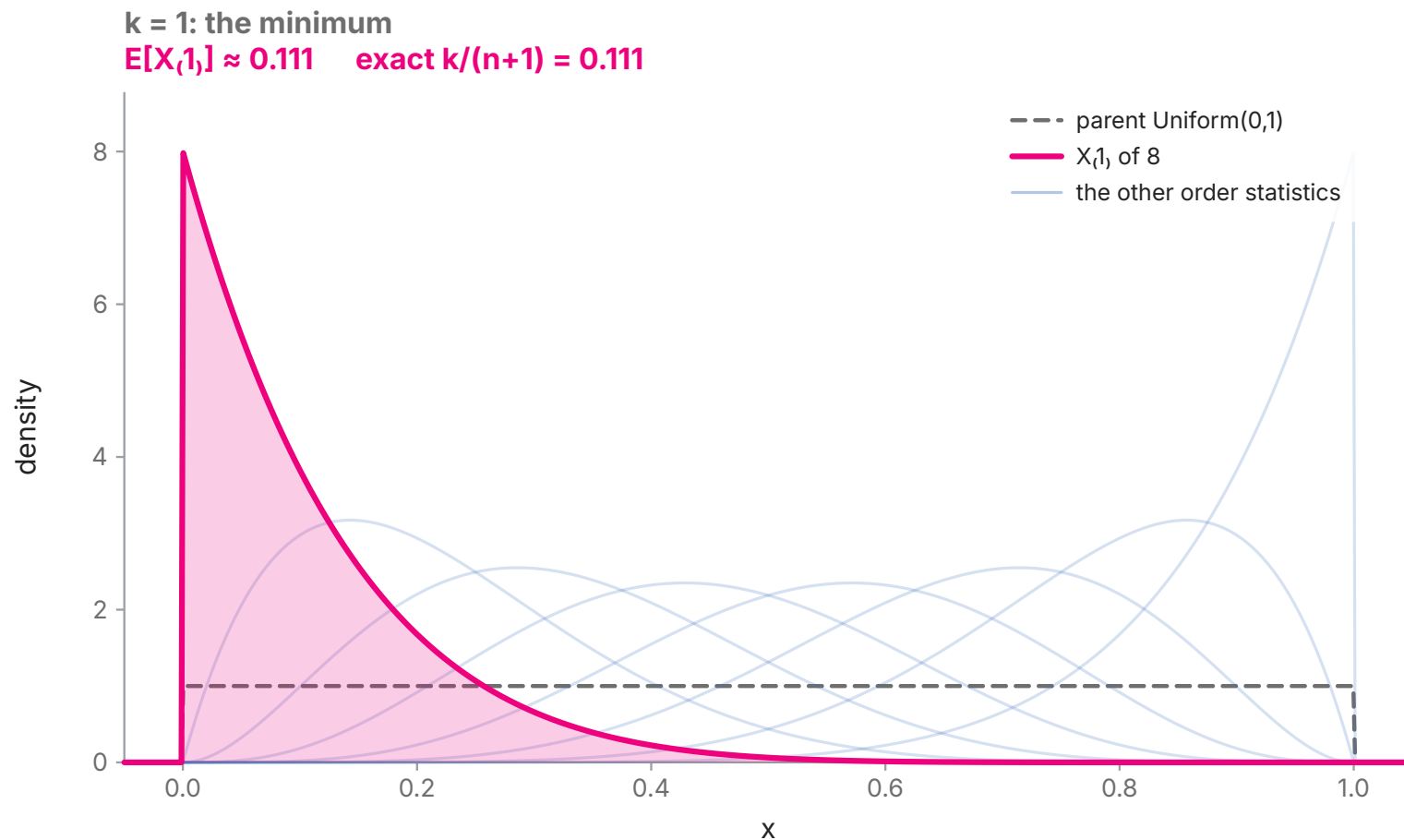
8

Which one, k

1

$k = 1$ is the minimum, $k = n$ the maximum.
Exact densities — no simulation.

Note how the extremes are the most skewed, and how they pull away from the parent as n grows.



Summary

Summary



Key Concepts

- Joint, marginal and conditional distributions
- Independence — pairwise versus mutual
- Covariance, correlation and the covariance matrix
- The multivariate normal distribution
- Transformations of random vectors
- Order statistics



Important Relationships

- Independence $\Rightarrow$ zero covariance, **but not conversely**
- For the multivariate normal: zero covariance $\Leftrightarrow$ independence
- Linear transformations preserve normality
- The joint determines the marginals, but not the other way round



The thread

Marginals describe each coordinate on its own. Everything interesting — dependence, conditioning, prediction — lives in what the marginals leave out.

Next Steps



Applications in Statistics

- Sampling distributions
- Maximum likelihood estimation
- Hypothesis testing for multiple parameters
- Regression analysis
- Principal component analysis
- Multivariate analysis of variance (MANOVA)



Further Study

- Characteristic functions
- Copulas
- Extreme value theory
- Time series analysis